

# Why the Proposed $p$ -adic Deutsch Principle Degenerates

## Sharp entropy bounds and the optimal Buzano substitute

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### Source questions

K. Mahesh Krishna,  *$p$ -adic Ghobber–Jamming Uncertainty Principle*, arXiv:2506.18913, Questions 3.7 and 3.9, asks for  $p$ -adic versions of the Deutsch entropy principle and the Buzano inequality. The source definitions and both questions are reproduced from PDF page 10 below.

*In particular, if  $x$  is supported on  $M$  in the expansion using basis  $(\{f_j\}_{j=1}^n, \{\tau_j\}_{j=1}^n)$  and  $x$  is supported on  $N$  in the expansion using basis  $(\{g_k\}_{k=1}^n, \{\omega_k\}_{k=1}^n)$ , then  $x = 0$ .*

We note that Inequality (5) will not give Inequality (2) as  $p$ -adic norm is not given using  $p$ -adic inner product.

We end the paper with an important open question. Let  $\mathcal{H}$  be a finite dimensional Hilbert space. Given an orthonormal basis  $\{\tau_j\}_{j=1}^n$  for  $\mathcal{H}$ , the **(finite) Shannon entropy** at a point  $h \in \mathcal{H}_r$  is defined as

$$S_r(h) := - \sum_{j=1}^n |\langle h, \tau_j \rangle|^2 \log |\langle h, \tau_j \rangle|^2 \geq 0,$$

where  $\mathcal{H}_r := \{h \in \mathcal{H} : \|h\| = 1, \langle h, \tau_j \rangle \neq 0, 1 \leq j \leq n\}$  [4]. In 1983, Deutsch derived following uncertainty principle for Shannon entropy which is fundamental to several developments in Mathematics and Physics [4].

**Theorem 3.6.** [4] (*Deutsch Uncertainty Principle*) Let  $\{\tau_j\}_{j=1}^n, \{\omega_j\}_{j=1}^n$  be two orthonormal bases for a finite dimensional Hilbert space  $\mathcal{H}$ . Then

$$(8) \quad 2 \log n \geq S_r(h) + S_\omega(h) \geq -2 \log \left( \frac{1 + \max_{1 \leq j, k \leq n} |\langle \tau_j, \omega_k \rangle|}{2} \right) \geq 0, \quad \forall h \in \mathcal{H}_r.$$

Let  $\{\tau_j\}_{j=1}^n$  be a  $p$ -adic orthonormal basis for a  $p$ -adic Hilbert space  $\mathcal{X}$ . Given a  $p$ -adic orthonormal basis  $\{\tau_j\}_{j=1}^n$  for  $\mathcal{X}$ , the **(finite)  $p$ -adic Shannon entropy** at a point  $x \in \mathcal{X}_r$  is defined as

$$S_r(x) := - \sum_{j=1}^n |\langle x, \tau_j \rangle|^2 \log |\langle x, \tau_j \rangle|^2 \geq 0,$$

where  $\mathcal{H}_r := \{x \in \mathcal{X} : \|x\| = 1, \langle x, \tau_j \rangle \neq 0, 1 \leq j \leq n\}$ . We then have the following problem.

**Question 3.7.** What is the  $p$ -adic version of Deutsch uncertainty principle (8)?

It has been observed very recently [11] that using Buzano inequality, (which we state below) one can give a simple proof of Inequality (8).

**Theorem 3.8.** [3, 5, 17] (*Buzano Inequality*) Let  $\mathcal{H}$  be a Hilbert space. Then

$$(9) \quad |\langle \tau, h \rangle \langle h, \omega \rangle| \leq \frac{\|h\|^2 (|\langle \tau, \omega \rangle| + \|\tau\| \|\omega\|)}{2}, \quad \forall h, \tau, \omega \in \mathcal{H}.$$

As Inequality (9) easily gives Inequality (8), we formulate following problem whose solution will give a  $p$ -adic version of Inequality (8).

**Question 3.9.** What is the  $p$ -adic version of Buzano Inequality (9)?

We note that there is an improvement of Inequality (8) due to Maassen and Uffink in 1988 [15] motivated from the conjecture by Kraus [10] made in 1987 (using Riesz-Thorin interpolation), whose  $p$ -adic version is also not known.

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### Resolution in one sentence

With the definitions in the source, the coherence of two  $p$ -adic orthonormal bases is *always one*; the Deutsch lower bound therefore collapses to zero, its classical upper bound is false, and the optimal universal Buzano estimate is merely the product of two Cauchy–Schwarz estimates. The exact replacement bounds are given next.

## Definitions

Let  $K$  be a non-Archimedean valued field and let  $X$  be an  $n$ -dimensional  $p$ -adic Hilbert space in the sense of the source. For an orthonormal basis  $\tau = (\tau_j)_{j=1}^n$  and  $\|x\| = 1$ , put

$$S_\tau(x) = - \sum_{j=1}^n |\langle x, \tau_j \rangle|^2 \log |\langle x, \tau_j \rangle|^2,$$

with  $0 \log 0 = 0$ . Define  $\psi(r) = -r^2 \log r^2$  and

$$C_K = \sup\{\psi(r) : r \in |K^\times|, 0 < r < 1\},$$

where the supremum of the empty set is zero.

**Theorem 1** (Sharp structural answer). *Let  $\tau$  and  $\omega$  be orthonormal bases of  $X$ .*

(i) *Their coherence satisfies*

$$\max_{1 \leq j, k \leq n} |\langle \tau_j, \omega_k \rangle| = 1.$$

(ii) *For every unit vector for which the entropies are defined,*

$$0 \leq S_\tau(x) + S_\omega(x) \leq 2(n-1)C_K \leq \frac{2(n-1)}{e}.$$

*For  $K = \mathbb{Q}_p$ , this becomes the exact universal estimate*

$$0 \leq S_\tau(x) + S_\omega(x) \leq \frac{4(n-1) \log p}{p^2}. \quad (1)$$

*Both endpoints in (1) are attained, even with  $\tau = \omega$ .*

(iii) *The optimal universal Buzano substitute is*

$$|\langle u, h \rangle \langle h, v \rangle| \leq \|u\| \|h\|^2 \|v\|. \quad (2)$$

*Its constant one remains sharp under the extra condition  $\langle u, v \rangle = 0$ .*

*Proof.* The norm Parseval formula stated in the source says

$$\|y\| = \max_j |\langle y, \tau_j \rangle|.$$

Every  $\omega_k$  has norm one. Hence, for each fixed  $k$ ,  $\max_j |\langle \omega_k, \tau_j \rangle| = 1$ , proving (i).

Write  $a_j = \langle x, \tau_j \rangle$ . Since  $\|x\| = 1$ , Parseval gives  $\max_j |a_j| = 1$ . Thus all coordinate moduli are at most one and at least one is one; the corresponding entropy term is zero. Consequently

$$0 \leq S_\tau(x) \leq (n-1)C_K,$$

and the same argument for  $\omega$  proves the first inequality in (ii). The elementary real-variable maximum  $\sup_{0 < r \leq 1} \psi(r) = 1/e$  proves the field-independent bound.

For  $K = \mathbb{Q}_p$ , every nonzero  $r \leq 1$  in the value group is  $r = p^{-m}$  for an integer  $m \geq 0$ . When  $m \geq 1$ ,

$$\psi(p^{-m}) = 2mp^{-2m} \log p.$$

The ratio of its value at  $m+1$  to its value at  $m$  is  $((m+1)/m)p^{-2} < 1$ . Therefore  $C_{\mathbb{Q}_p} = 2p^{-2} \log p$ , which gives (1). In standard  $\mathbb{Q}_p^n$ , take both bases canonical. The unit vectors

$$x = (1, p, \dots, p) \quad \text{and} \quad x = (1, 1, \dots, 1)$$

attain respectively the upper and lower endpoints; all their coordinates are nonzero, as required by the source's domain.

Finally, the defining Cauchy–Schwarz axiom gives

$$|\langle u, h \rangle| \leq \|u\| \|h\|, \quad |\langle h, v \rangle| \leq \|h\| \|v\|,$$

and multiplication yields (2). In standard  $K^2$ , set  $u = e_1$ ,  $v = e_2$ , and  $h = e_1 + e_2$ . Then  $\langle u, v \rangle = 0$ , while both sides of (2) equal one. No universal factor smaller than one—even at zero overlap—is possible.  $\square$

## Consequences for the two questions

Part (i) makes Deutsch’s proposed overlap expression

$$-2 \log \left( \frac{1 + \max_{j,k} |\langle \tau_j, \omega_k \rangle|}{2} \right)$$

identically zero. This is sharp as a universal lower bound by the all-unit example above. The classical upper bound  $2 \log n$  also does not survive: over  $\mathbb{Q}_2^8$ , with equal canonical bases and  $x = (1, 2, \dots, 2)$ ,

$$S_\tau(x) + S_\omega(x) = 7 \log 2 > 6 \log 2 = 2 \log 8. \quad (3)$$

Thus (1), not the classical interval, is the sharp answer for the entropy defined in Question 3.7.

For Question 3.9, (2) is optimal. The orthogonal equality example shows precisely why a Buzano improvement depending on  $|\langle u, v \rangle|$  cannot exist in this category. It therefore cannot supply a nonzero entropic uncertainty term.

## Audit and scope

The accompanying exact-arithmetic script verifies the discrete entropy optimization, (3), and the orthogonal equality example. A bounded search of the run indexes, exact question wordings, arXiv, and the author’s adjacent  $p$ -adic uncertainty papers found no later answer. Novelty confidence is moderate pending specialist review. The conclusion concerns exactly the entropy and orthonormal-basis definitions printed in the source; alternative probability normalizations may lead to different theories.